

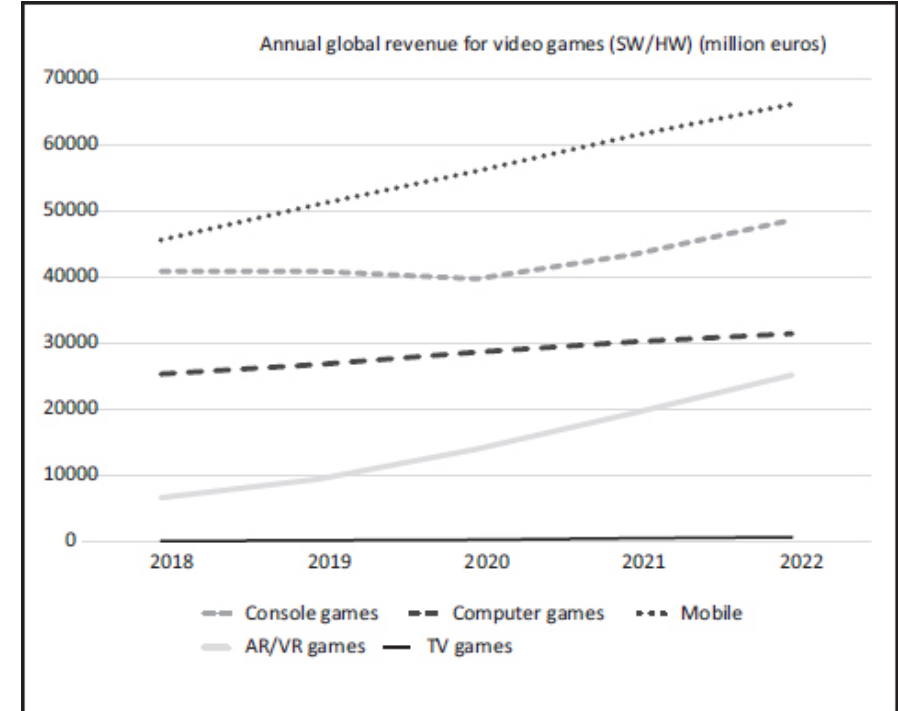
# Video Game Programming: Game Project Management

L05

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# Game market

- Game market has an annual global revenue of several billion euros
- Game market is growing and will likely continue to grow in the next
- Until 10-15 years ago, game market was largely impacted by console cycles (new console, burst of revenue from new games, revenue slow decrease, new consoles within 5 years)
- Major console manufacturers profit is in games, because developers pay licenses
- Today, longer console cycles and less console impact because of new markets with different trends



# Game industry

- Income by selling to consumers (retail or download), subscription fees, advertising, in-app purchases (e.g., in freemium), and royalties
- Game industry is complex, with specializations, big budgets, new technologies
- AAA games (higher budgets) are made by highly trained people in big companies
- Trend towards larger production teams (e.g., SimCity: 20 persons, Halo 2: 80, Call of Duty MW 2: 200, Grand Theft Auto 5: 500), increased development costs, tougher competition
- Specialized experts, focusing on different aspects of sound, programming, animation, graphics, marketing, game design, and production
- Team size varies with business model and platform (small for mobile, medium for social, large for online, huge for AAA), with variations within platform
- Need for game project management to coordinate teams and meet deadlines

# Independent game development

- Between 1990 and 2000, developing and publishing a game all alone was not realistic
- As budgets and teams grew, in the indie movement, small teams started developing simpler games on their own
- Several factor made this possible:
  - New platforms and new business models (e.g., mobile app stores, Steam greenlight / direct)
  - Indie access to consoles
  - New tools such as Unity to “democratize game development”
  - Crowdfunding (e.g., Kickstarter)
- Low-budget games (e.g., Darwinia, Limbo, Minecraft) become mainstream online or on mobile
- Indie game development needs project management to meet strict deadlines of game jams, and to complete the project

# Project management

- It consists of a set of tools, practices, methodologies, and skills that help projects thrive
- It supports communication, clarity, collaboration
- It can start with a Project Charter Form (in fig.)

## PROJECT CHARTER

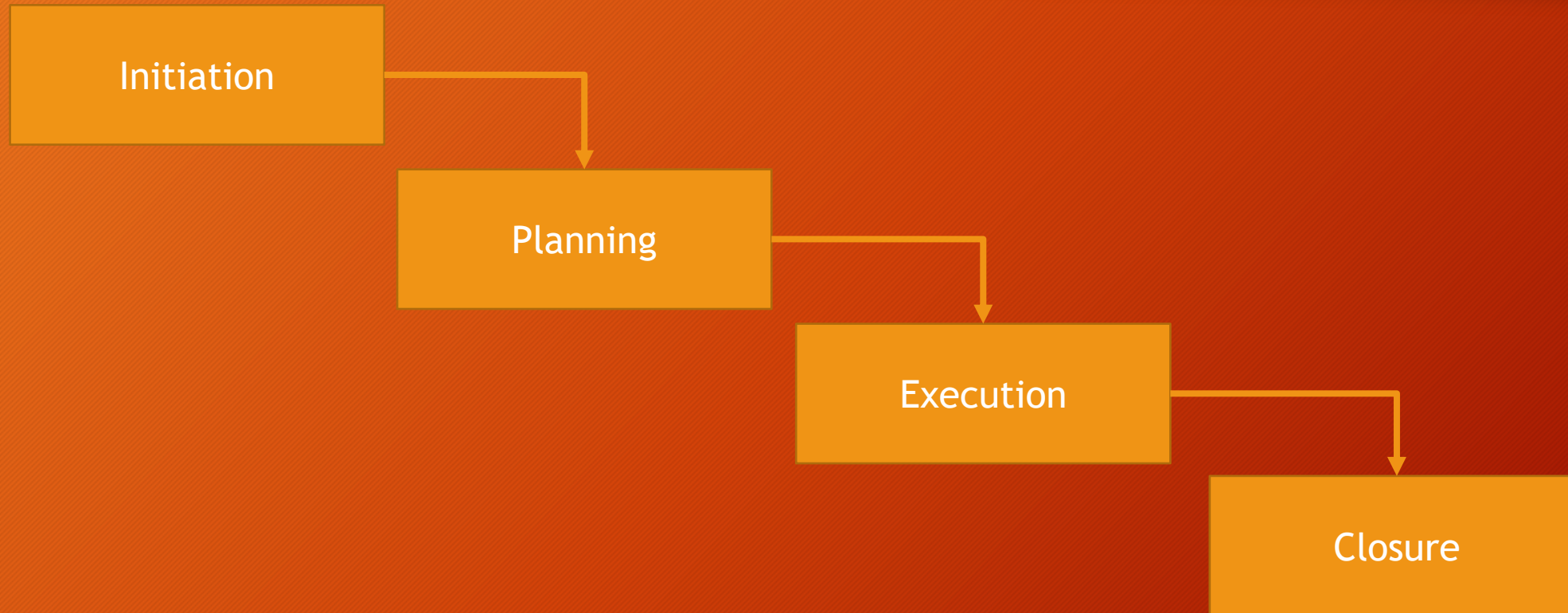
### Project Charter

|                         |  |
|-------------------------|--|
| Project Name            |  |
| Project Description     |  |
| Project Sponsor         |  |
| Project Team Member(s)  |  |
| Project Objectives      |  |
| Project Milestones      |  |
| Assumptions             |  |
| Project Completion Date |  |

We agree that this is a viable project. The Project Team Member(s) are authorized to begin the planning, design, and development process for this project.

|                        |  |
|------------------------|--|
| Project Sponsor        |  |
| Project Team / Manager |  |

# The four phases in waterfall model



# The four phases for game projects

- Different naming for slightly different phases:
  - Conceptual, Design, Production, Testing (Egenfeldt-Nielsen, Smith, Tosca, Understanding Video Games, 2024)
  - Conceptualize, Prototype, Test, Analyze (Unity curricular framework)
  - Analysis, Design, Implementation, Testing (Gibson Bond, Introduction to Game Design, Prototyping, and Development, 2023)
- All share the importance of testing, which is sometimes included in execution or closure or added in-between in the general model

# Analysis/Conceptualize phase

- Analysis is about understanding for whom you are designing the game, involved resources, and fastest path to a demo
- Game concept is formulated, in a few pages, to convey the core idea of game and features, such as platform and concept art
- Designer creates a more detailed game proposal, covering market analysis, technical issues, budget projections, audiovisual style and descriptions of how the game would feel to play, to attract funding and to plan the production

# Design phase

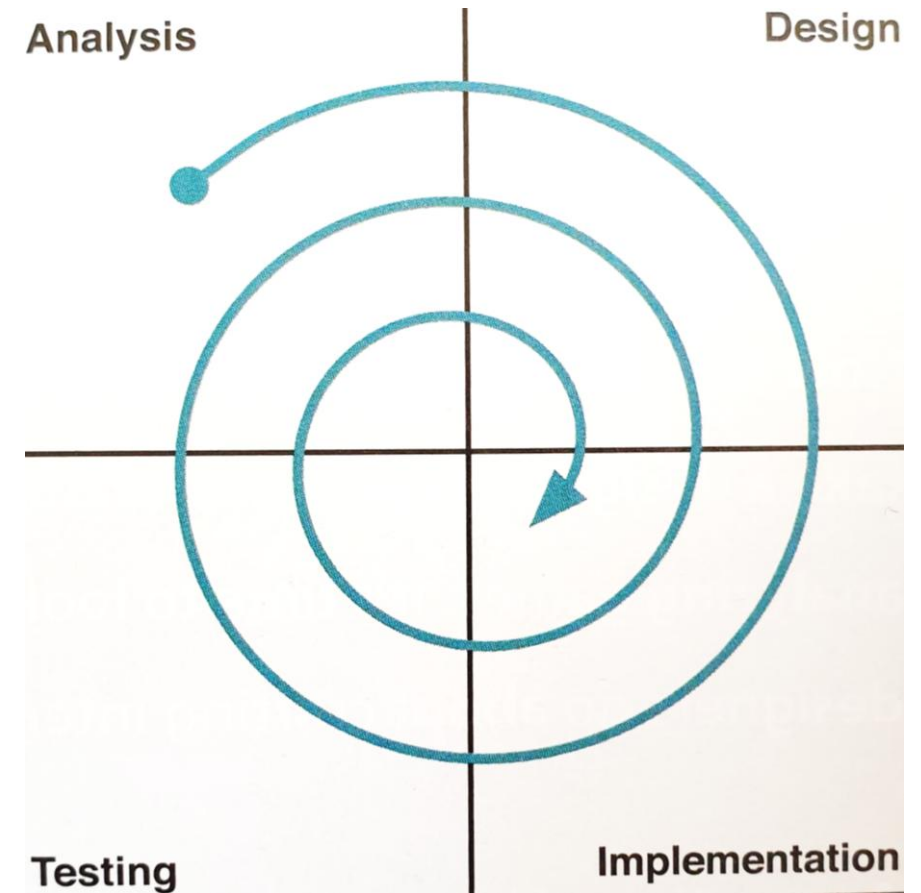
- Starts with brainstorming and listening to audience, team, client, game, and yourself
- Game designers describe functional and technical specifications with concrete details about features and how the player will interact with the game, specifying how to achieve the desired design on a technical level
- The result is a design document (up to some hundred pages) with text, illustrations, mockups, concept drawings, and other details, such as lists of game objects
- Design document is updated constantly as development advances
- Based on the design document, the developer decides the game engine and the tools to handle assets such as music and textures
- Next step is the creation of a working prototype to give a feel for the game, ensure the core game design works, and get funding

# Production/implementation and test phases

- With design document and prototype, developers start game production
- Developers address unexpected challenges and may be forced to change game elements during production
- Production involves writing code and creating assets
- When separate elements are made, they are linked in the alpha version, which contains all main elements for a playable game but lacks fine tuning
- Alpha is used internally to test for errors and issues in usability and playability
- These problems are addressed before the beta for real-world testing, which testers play for a long period, reporting problems and giving suggestions
- Based on input from beta testers, final changes and corrections are made to make the gold master for the actual release of the game
- In the past, gold master could mean the end of developers' tasks, while now they are usually involved in releasing patches, add-ons, DLCs...

# Rapid iteration and prototyping

- Game design is 1% inspiration and 99% iteration (Swain)
- Waterfall model is replaced by iterative design, repeating iterate-test cycle
- Developers build prototypes, test them, refine them, and then test them again
- Tests should have one or a few clear issues to address
- When developers refine prototypes, they should change only a few things at a time



Iterative design process (Fullerton, Swain, Hoffman, Game Design Workshop, 2008)

# 3D game engines

- 3D engines allow developers to build and display 3D environments without writing low-level OpenGL / DirectX / Vulkan code
- Game engines are sets of tools that make the game design and development process easier
- 3D game engines display 3D environments and provide additional tools for game development
- Engine is the software that provides the basic architecture but not the concrete content
- Game engines can handle artificial intelligence, audio, physics, UI,...
- Popular 3D game engines include Unreal Engine, CryEngine, Godot, Unity3D,...

# Game assets

- Think about a scene from a game you like and list every asset in it
  - 3D objects
  - Images
  - Text
  - Lighting
  - Animations
  - Effect sounds
  - Music
  - Code
- How long does it take to make each asset?
- Which tools can we use?

# Asset development tools

- Graphic file formats (.jpg, .gif, .png) and editors (Photoshop, Gimp, Paint)
- Video file formats (.mp4, .avi, .mov) and editors (Premiere, Da Vinci Resolve, Vegas)
- Sound file formats (.wav, .ogg, .mp3) and editors (Audacity, Audition, GarageBand)
- 3D file formats (.max, .3ds, .fbx) and editors (3ds Max, Maya, Blender)
- Source code and libraries file formats (.cs, .dll, .jar) and editors (Notepad++, Visual Studio, Rider)

# Make or buy

- To have the required assets two strategies are available
- Make, i.e., create the assets internally
  - Full control on design and changes
  - No issues with license
  - No need for money to buy
- Buy, i.e., use third-party assets
  - Speeds up development saving lots of time
  - Already tested assets, improved over time
  - Sometimes very affordable or even free

# Versioning code (and assets)

- Version control is fundamental to keep track of changes and synchronize work of team
- Different systems can be used (e.g., Subversion, GIT)
- Some websites allow for free hosting (e.g., GitHub, Azure DevOps)
- There are several clients for SVN and GIT (text-based, graphical, integrated in code editors)
- Perfect for code, useful also for other assets
- In Unity, it works better if assets are serialized as text

# Asset management in Unity

- All (or almost all) assets required for game development can be imported in Unity
- A good asset management strategy is needed
  - Should assets be grouped by type?
  - Should nested folders be used?
- Unity automatically reimports assets on change
- Changes are applied even while running
- It is better to use native (not exported) resource files (with some exceptions...)

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